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The Editors
Methodology and Computing in Applied Probability

Dear Editor,

I submit the manuscript "Two end modes: logarithmic asymptotics and importance sampling for the peak load of a pickup-and-delivery route" for consideration in *Methodology and Computing in Applied Probability*.

The paper studies the tail of the peak of a load process that arises in vehicle routing: a maximum of sums of gamma variables coupled by one-factor Gaussian copulas. It proves logarithmic asymptotics with explicit spectral scales, a sharp threshold for the efficiency of single-shift importance sampling, an exact description of how the single shift underestimates without inflating its own reported variance, and a sufficient condition, reduced to a computable covering constant, for the logarithmic efficiency of a mixture of shifts. A short numerical section checks the conditions and predictions on routes of two regional plans.

In the interest of transparency: an earlier, computation-oriented version of this work was submitted to the *INFORMS Journal on Computing* (JOC-2026-10-OA-2667). The area editor, Prof. Shane Henderson, declined it without review because the model is narrow, noted that the analytical treatment of the importance-sampling problem was its valuable part, and recommended a journal devoted to probability. The present manuscript has been rewritten around that analysis, which combines rare-event asymptotics with the design and efficiency of simulation estimators and so falls within the scope of this journal: all proofs are complete (the earlier version gave sketches), the results on the silent underestimation and on mixture efficiency are new, and the computational material has been reduced to an illustration. I intend to publish under the journal's standard (non-open-access) model.

The manuscript is not under consideration elsewhere. Code and data are publicly available. There are no competing interests and no funding to declare.

Sincerely,

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